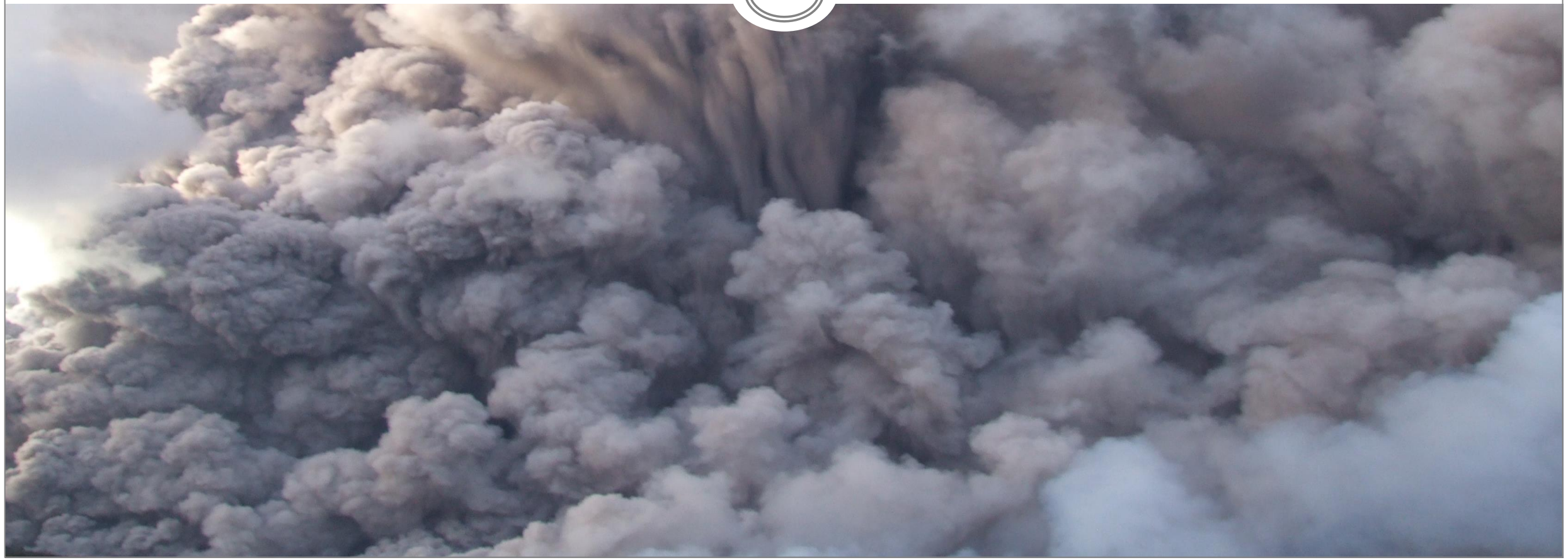
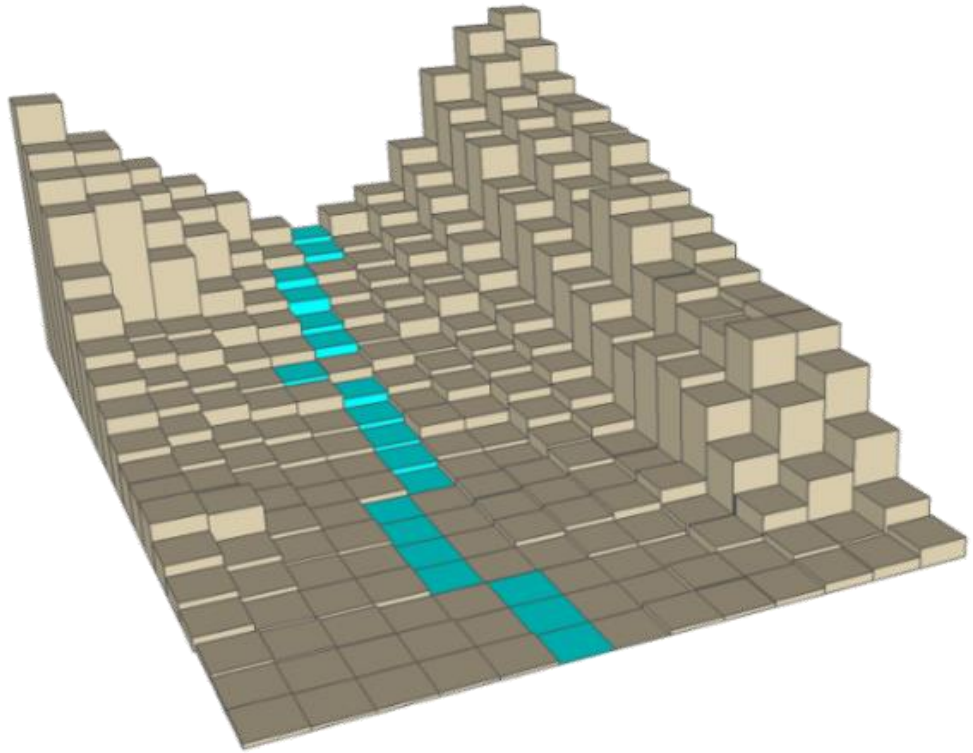
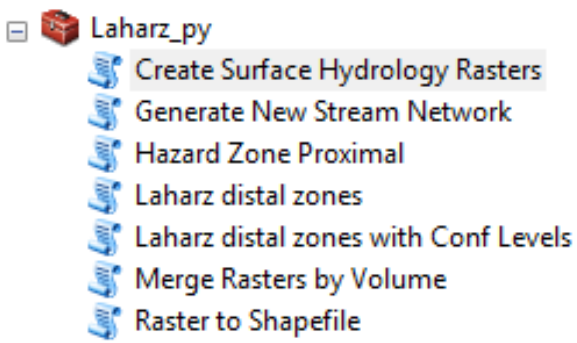
Stream networks and H/L ratio: more details

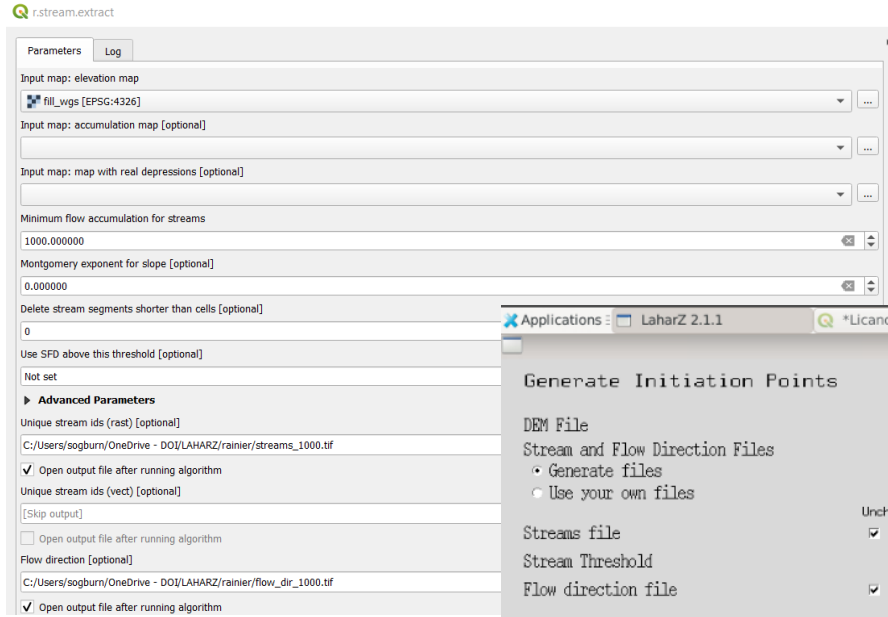


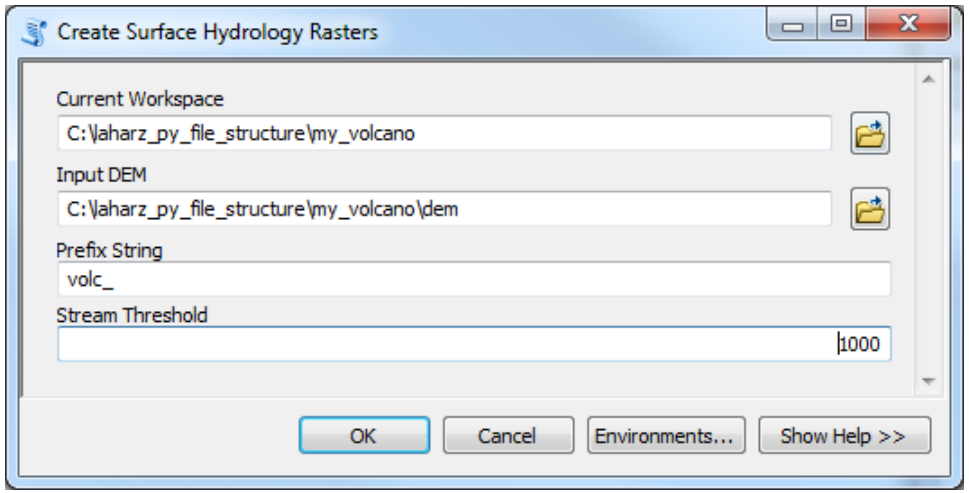
Stream Network

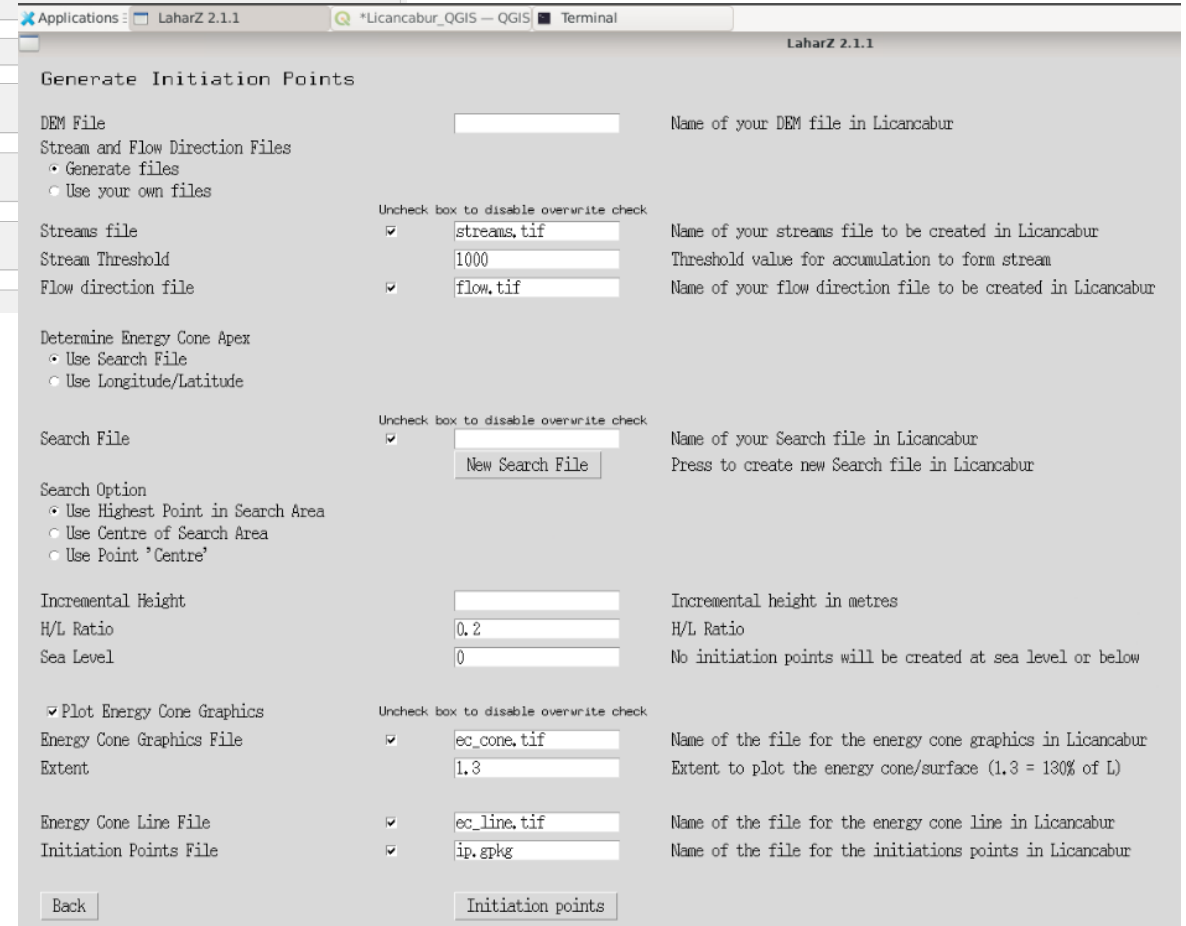
- “Create surface hydrology rasters” and “Generate Stream Network” tools in ArcGIS LAHARZ, the “r.stream.extract” in QGIS, or the newest version of the standalone Python LAHARZ generates stream network using the DEM, and a user defined stream threshold
- This is the number of upstream cells that have to flow into a cell to consider it a stream cell. A smaller number here will make a stream network with more branches and longer streams, a higher number creates a simpler stream network with fewer branches
- The network should generally match published maps, but should represent the *current* stream network



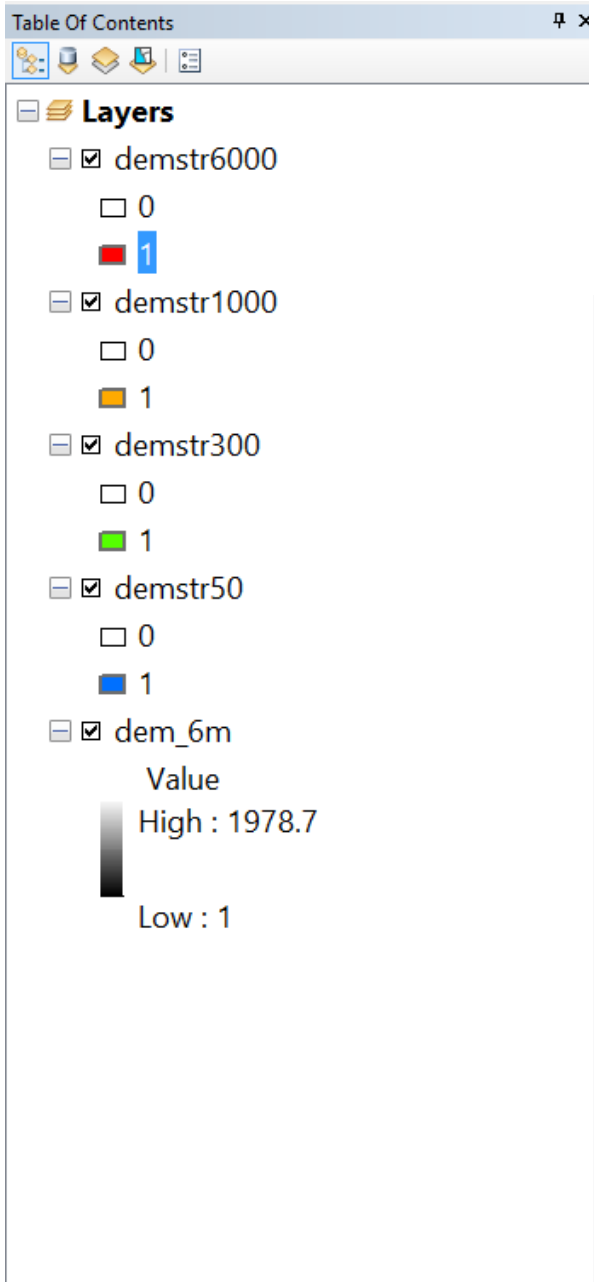






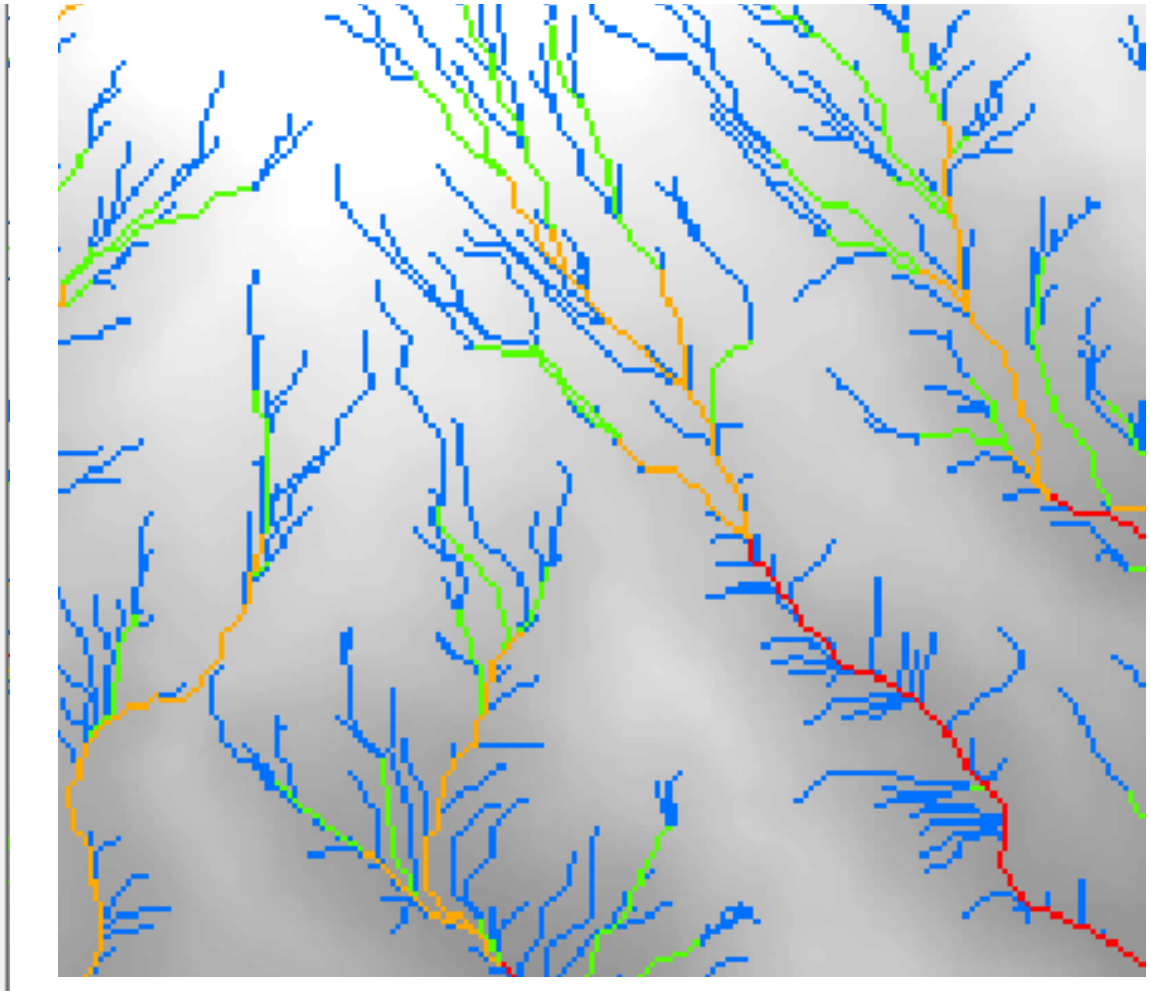


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High stream threshold: shorter stream network with fewer branches

Low stream threshold: longer stream network with more branches



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78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

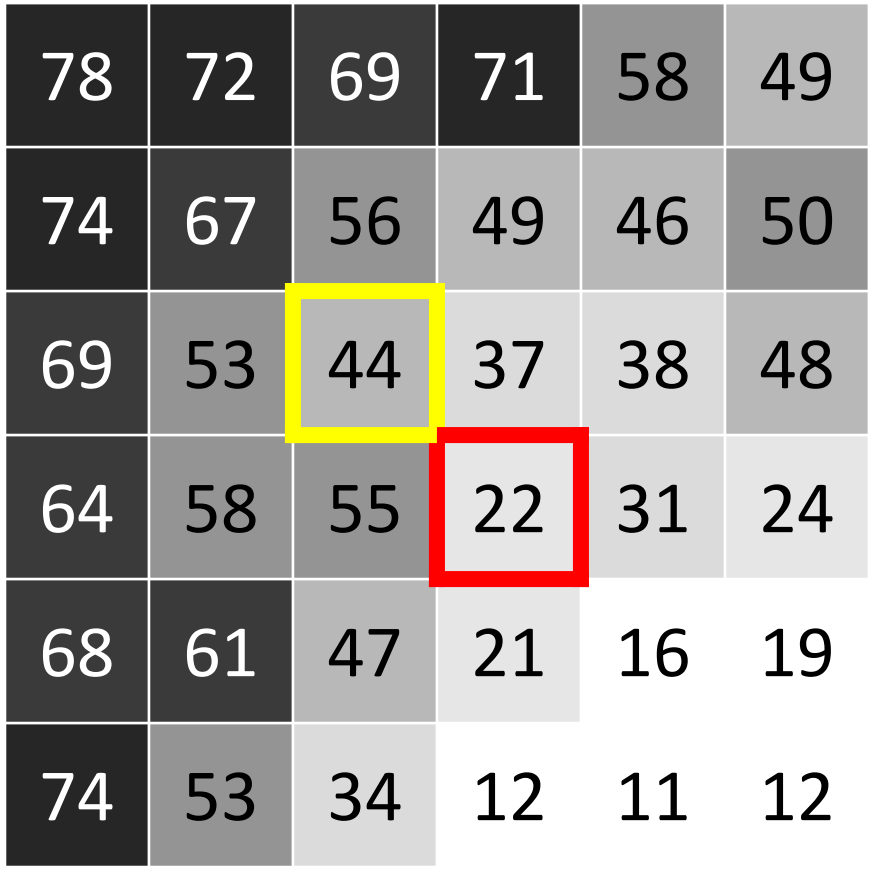
Elevation (DEM)

78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

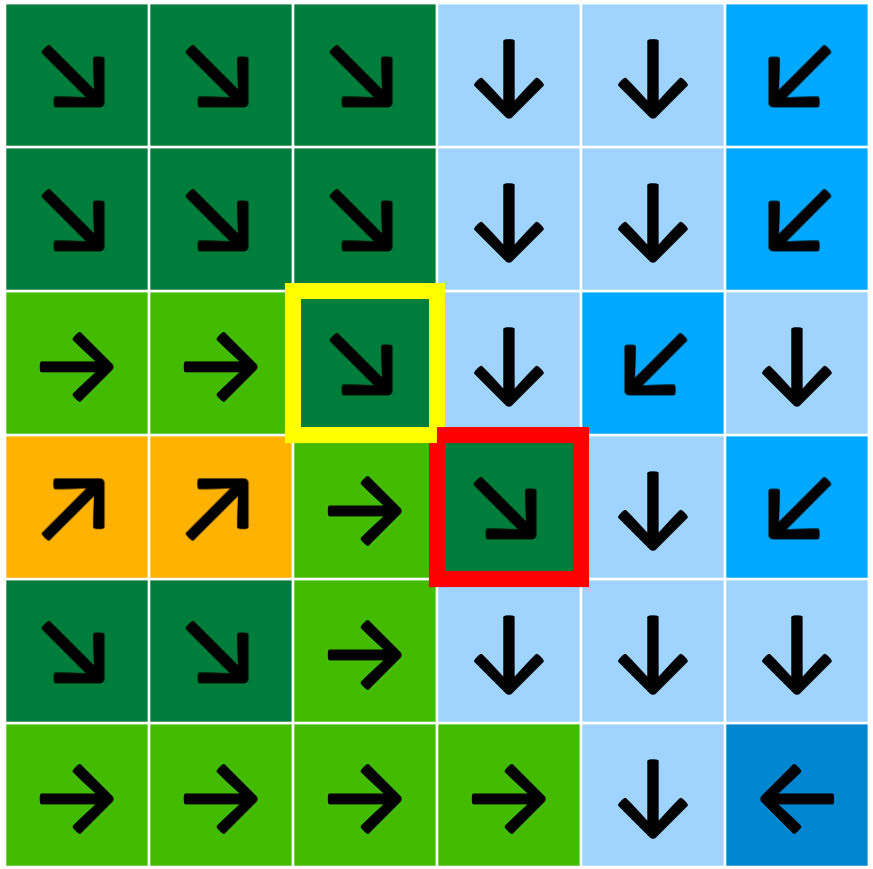
Elevation (DEM)

Elevation→Flow direction

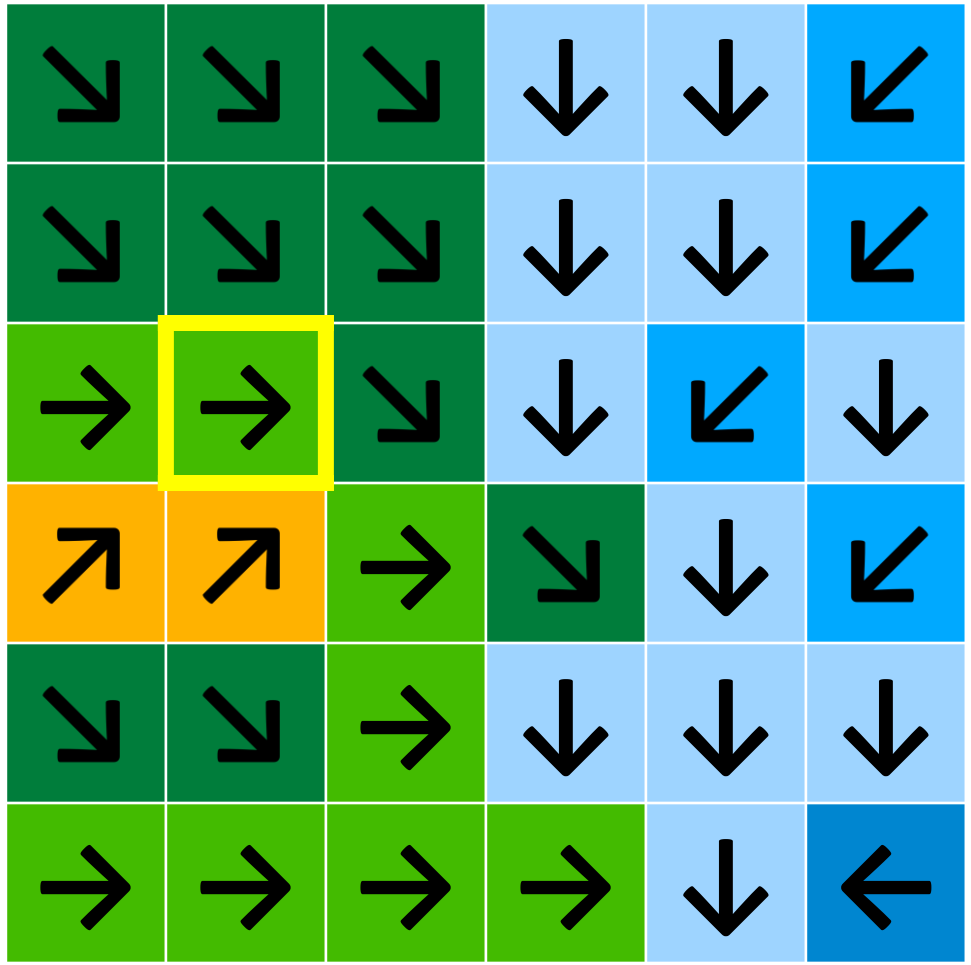
Where would a drop of water that fell on the yellow square go?



Elevation (DEM)



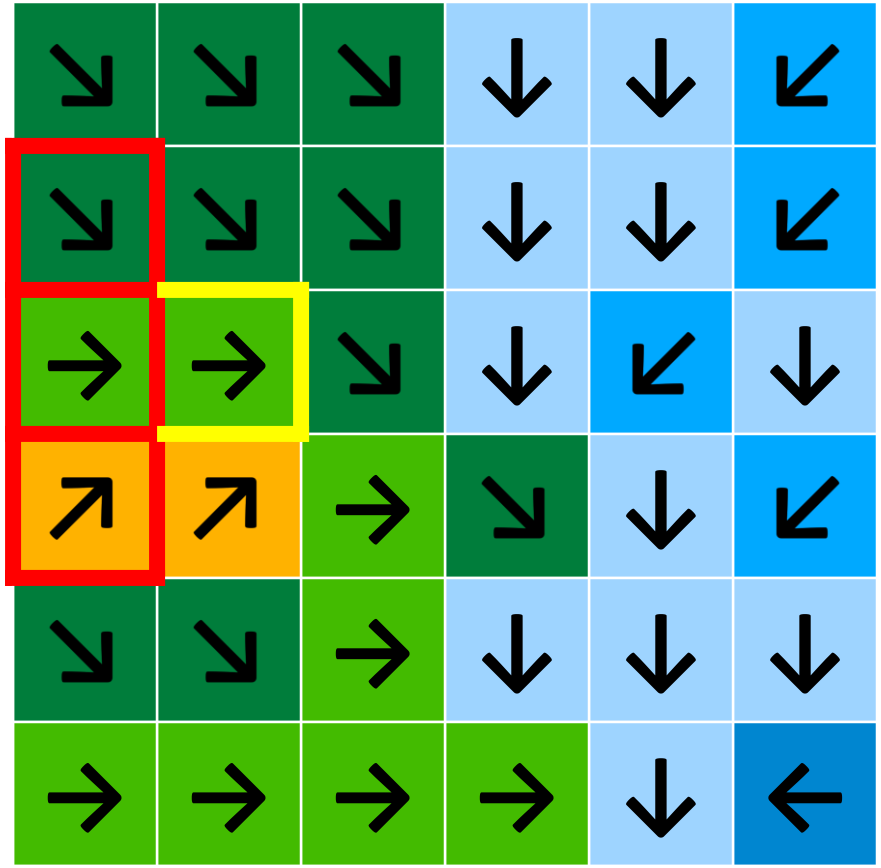
Flow direction



Flow direction

Flow direction → Flow accumulation

How many cells flow into the yellow square?



Flow direction

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

Flow accumulation → Stream network

With a stream threshold of 2 what does the stream network look like?

Stream threshold = 2

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0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 2

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

Stream threshold = 6

Flow accumulation → Stream network

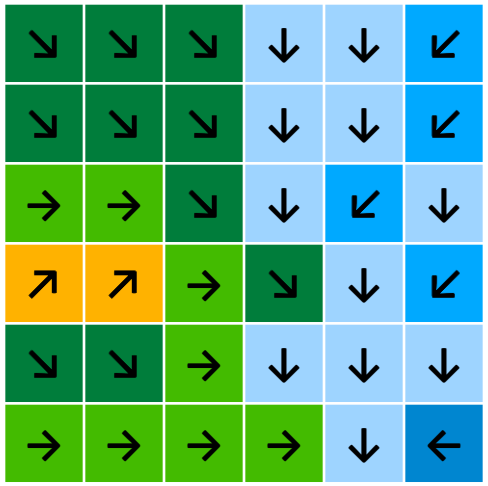
With a stream threshold of 6 what does the stream network look like?

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 6

78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

Elevation (DEM)



Flow direction

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 2

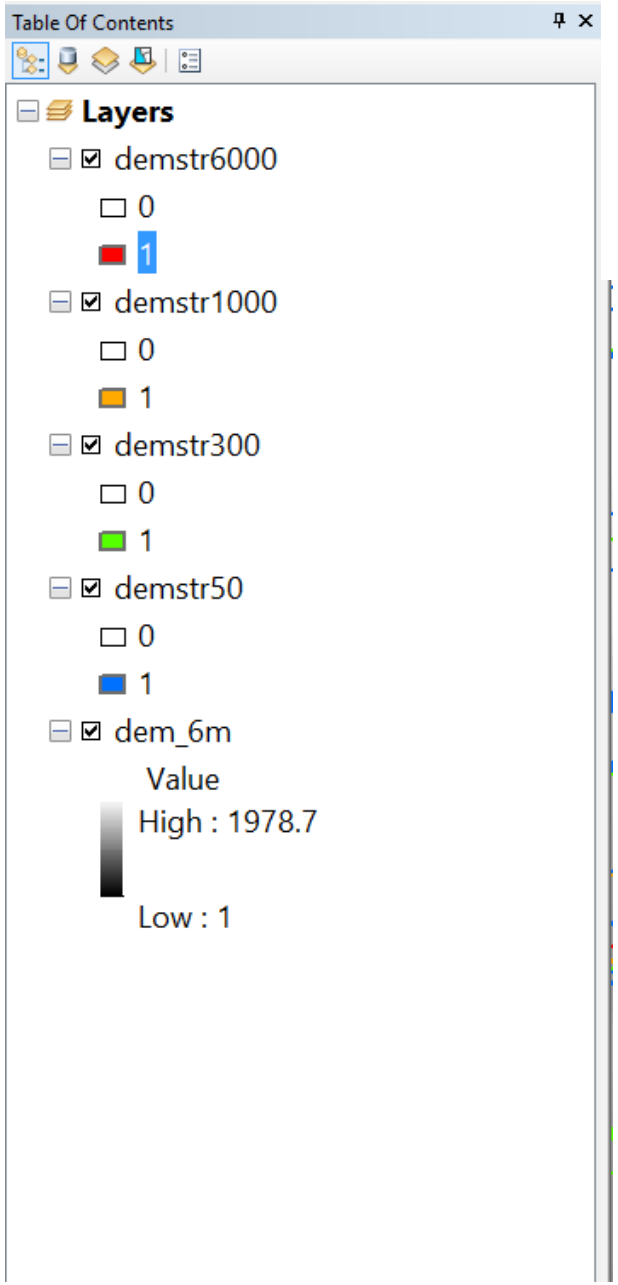
0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 6

Lower stream threshold: longer stream network with more branches

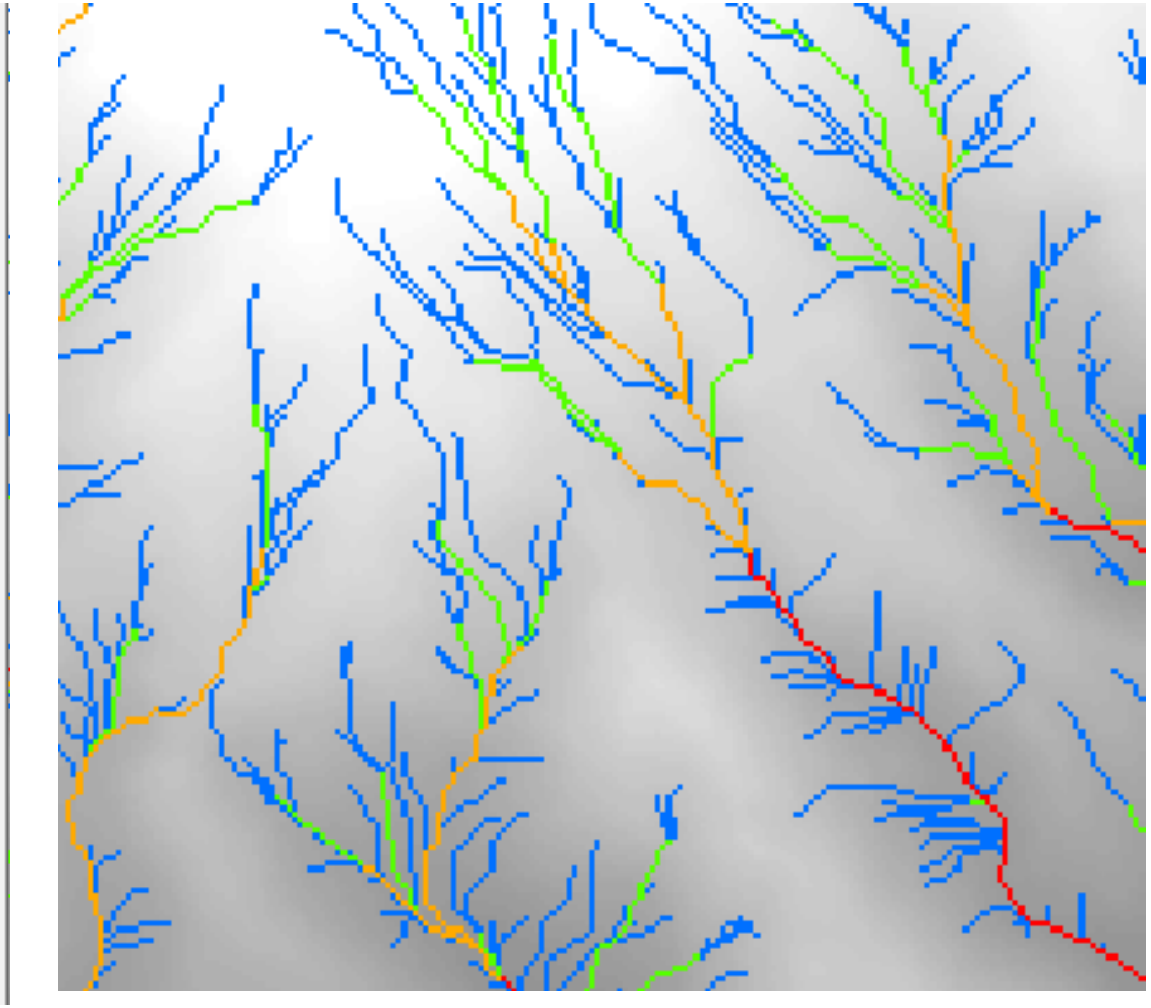
Higher stream threshold: shorter stream network with fewer branches

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Lower stream threshold: longer stream network with more branches

Higher stream threshold: shorter stream network with fewer branches



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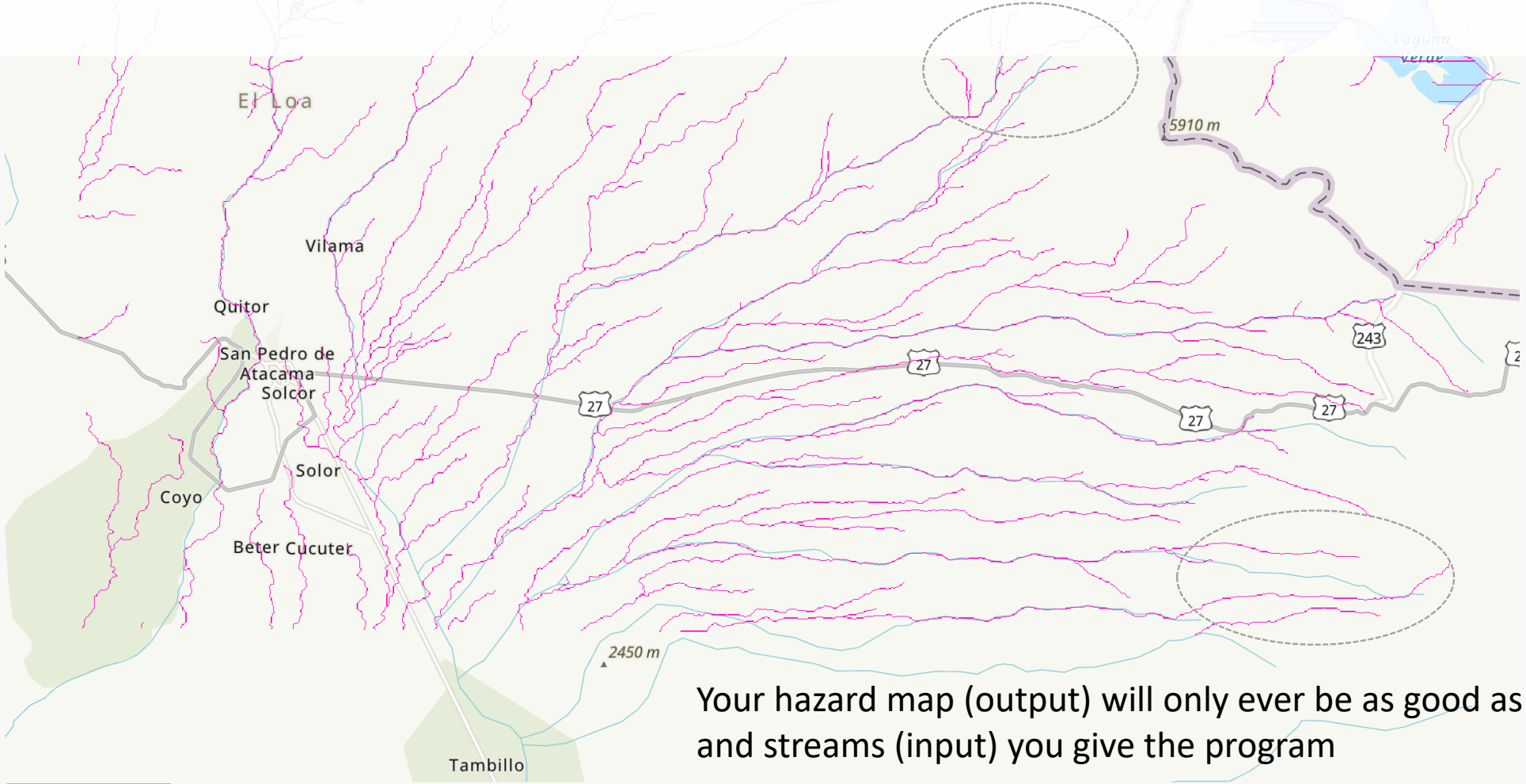
How to decide on the stream threshold

- Compare your stream network to a recent topographic map
- Compare your stream network to a recent satellite image
- Compare your stream network to field measurements
- Examine the stream network in locations where lahars/flash floods have been known to spill over into other channels

Evaluate the computer-generated streams to a basemap or (better) an authoritative map made locally and recently.

Reasons why there may be differences:

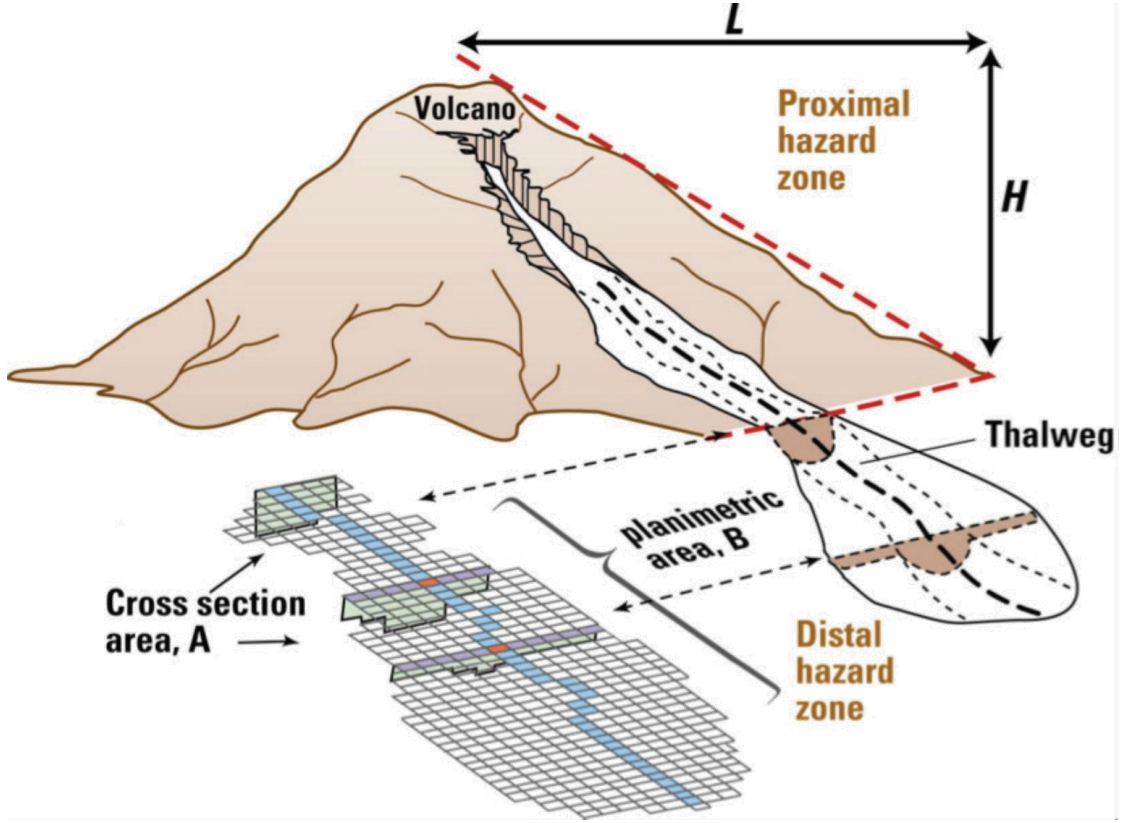
- Landscapes change over time, seasonally or catastrophically
- Man-made features like paving, canals, dams or bridges may alter the topography
- Map made at the wrong scale to be accurate at this local- regional scale
- DEM may be older and may not reflect changes
- Errors



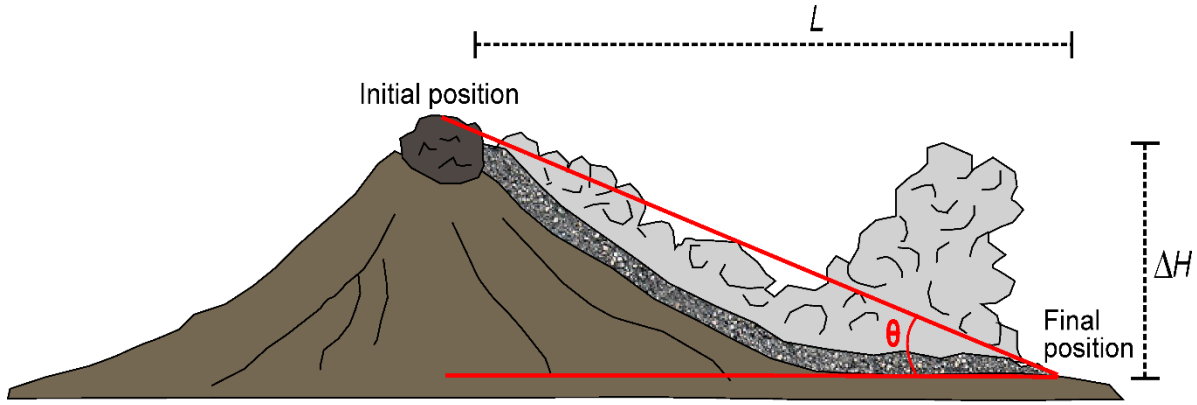
Your hazard map (output) will only ever be as good as the DEM and streams (input) you give the program

ΔH/L ratio and energy cones

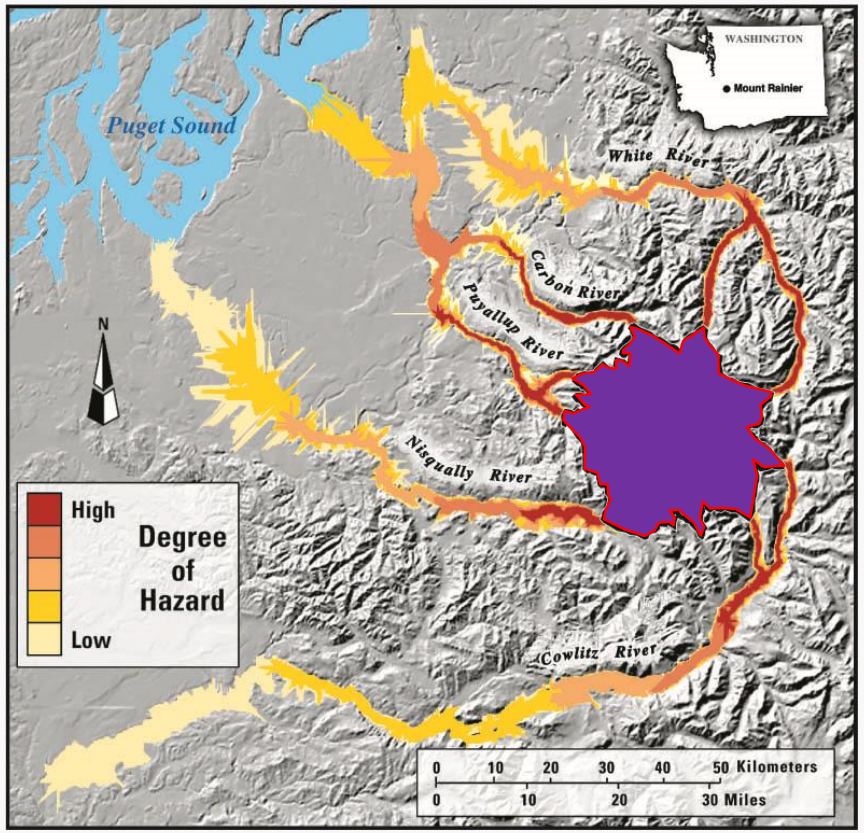
For LAHARZ



For PDCs



H/L = energy cone, coefficient of friction, Heim coefficient, H/L ratio



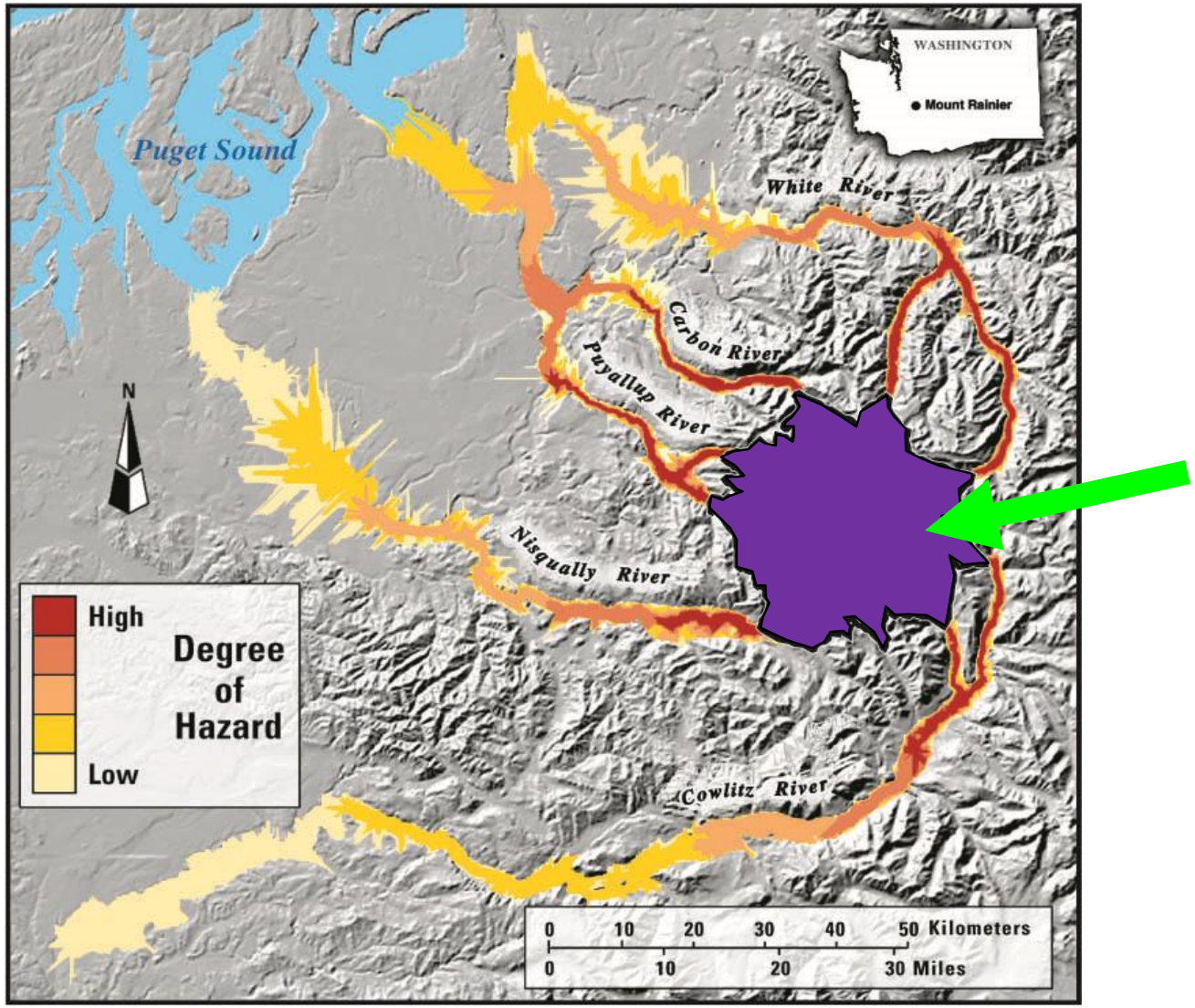


On a hazard map, the Proximal Hazard Zone, or highest hazard zone, is the most hazardous, most destructive area around a volcano because of steep slopes and proximity to the summit.

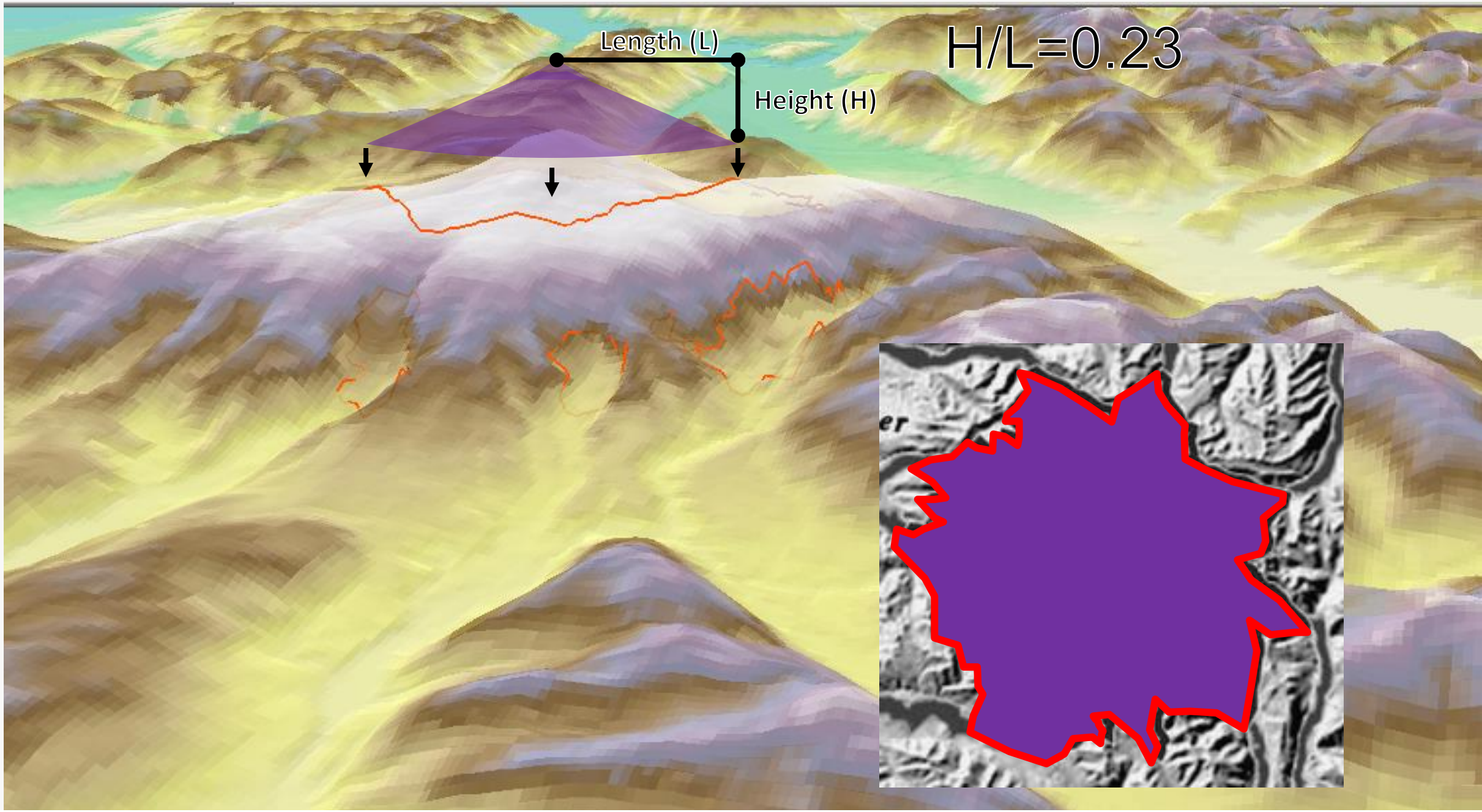
Proximal hazard zones usually include pyroclastic flows, volcanic bombs, vent lava flows and tephra fall.

For LAHARZ, creating a proximal hazard zone is also a convenient way to determine start locations for modeling lahars.

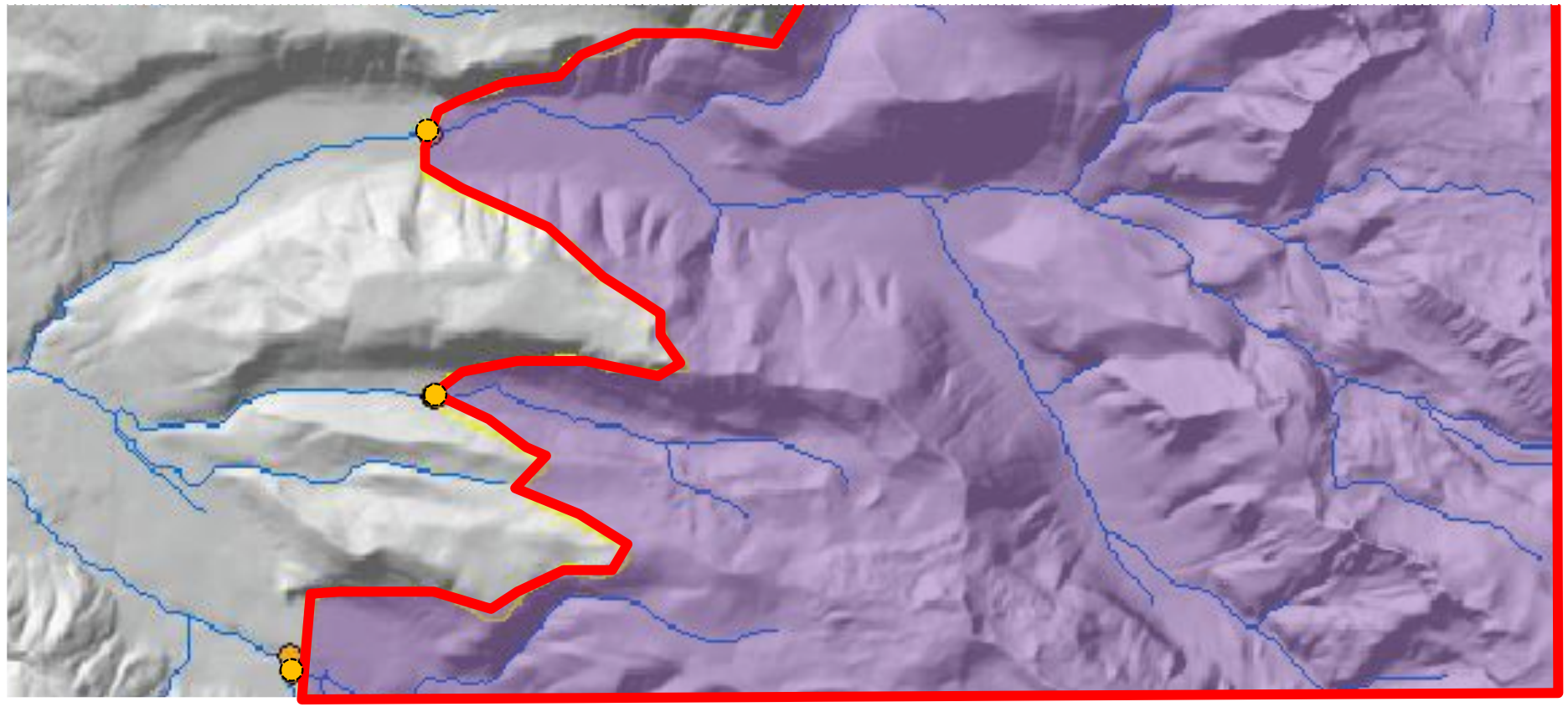
Lahar hazard zones computed for Mt. Rainier with proximal hazard zone defined by *H/L* = 0.23



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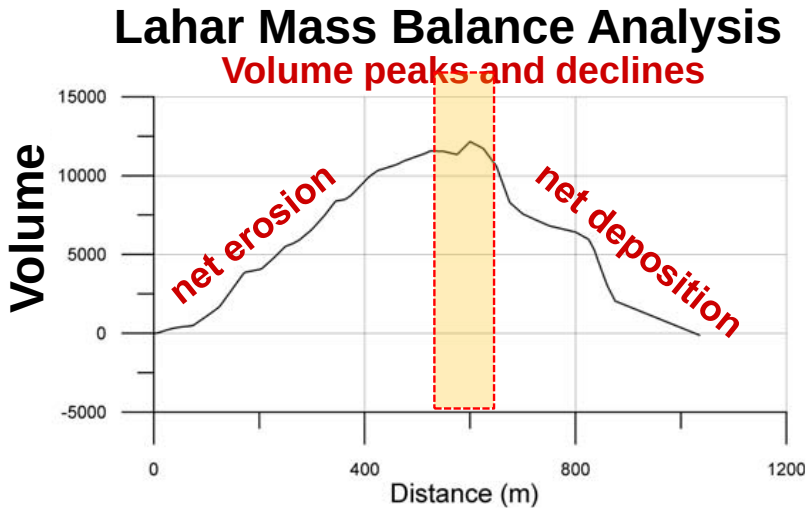
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Proximal Hazard Zone for LAHARZ

Why use H/L Cone for lahars?

- Lahars may initiate anywhere on the volcano edifice. We can't know an exact start location in advance.
- Loose pyroclastic flow deposits and ashfall are often the source material for lahars and are most abundant within the H/L cone.
- Lahars gain volume (entrain material) on steep slopes.



- So, we start modeling lahars by estimating where on the flanks the flow volume is greatest.
- We do this by defining an H/L cone that marks the transition between erosion and deposition. That is, inside the H/L cone, the volume is still increasing. Outside the H/L cone, the volume is maintained or decreasing (depositing).
- How do we identify an H/L cone that represents this transition?
    - from net erosion to net deposition
    - from steep slopes to moderate slopes (20°-30°), that is, find the break in slope
    - from pyroclastic debris fans to more channelized flows

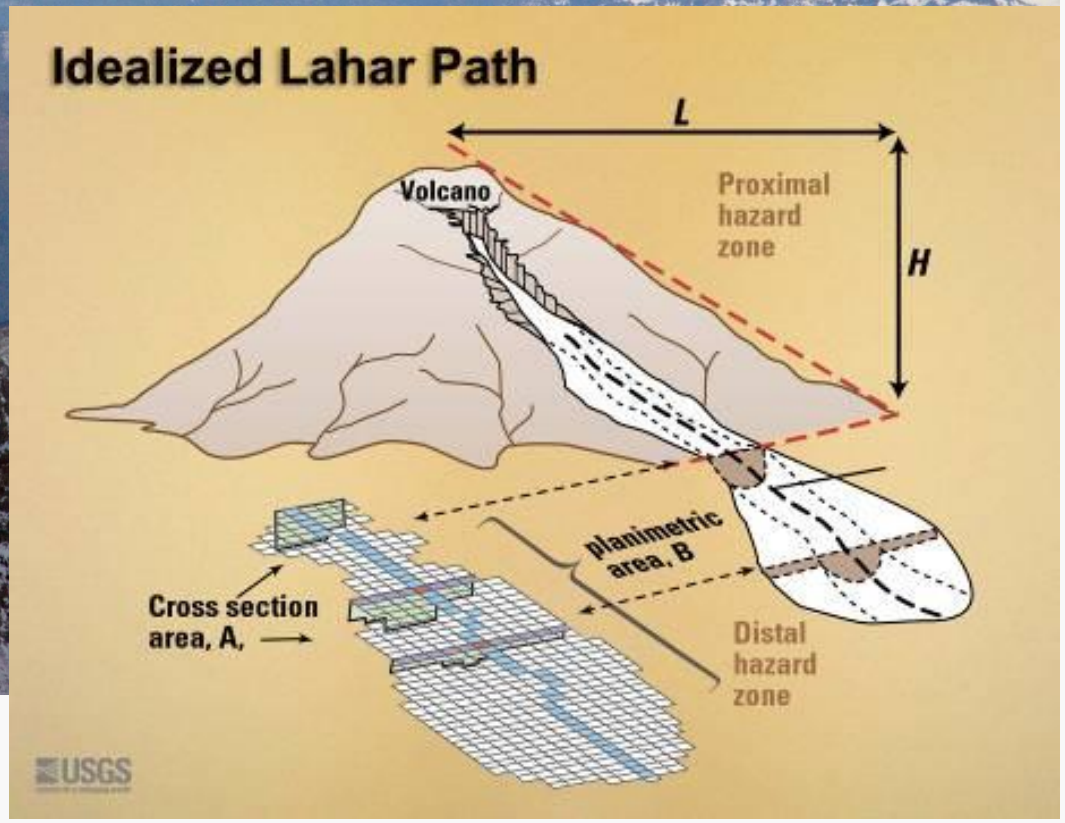
Mt. Rainier, west side

L

V

H

Begin flow hazard-zone computation where deposition first exceeds erosion



Okay, so how do we choose the H/L value?

PDC deposit fans are useful for choosing an H/L cone/proximal hazard zone

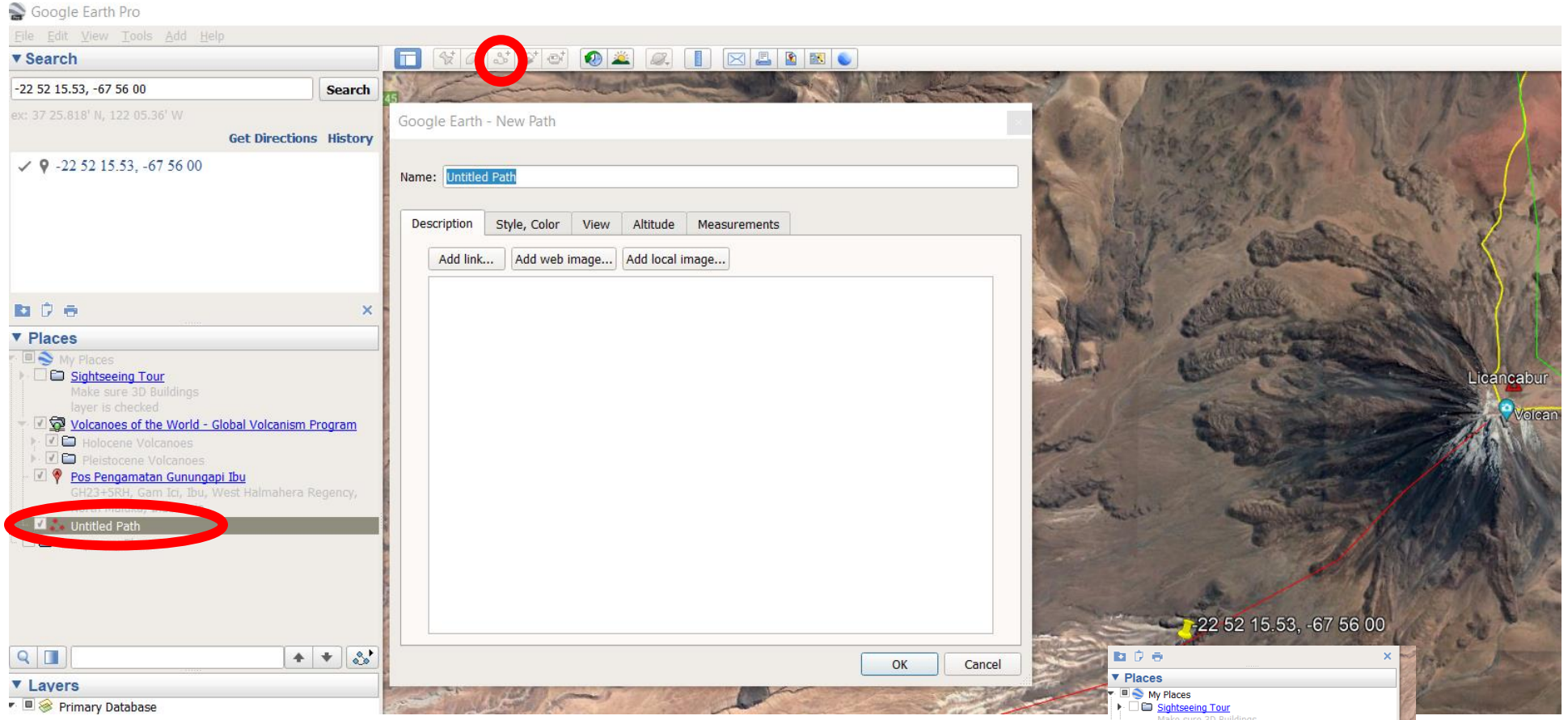
- **Visualize volcano in 3-D:**
  - Find photos (Google image search, GVP)
  - Use Google Earth to get oblique views, fly around, and make a topographic profile.
- **Examine volcano records of past events:**
  - geologic maps; observatory logs; GVP info
- **Look at the geomorphology:**
  - landforms, farming, land use
- **Look for PDC units in the field**
- **Examine satellite imagery**
- **Use global record of analog volcanoes (PDC H/L values)**
- **Start with H/L of 0.3 and refine**

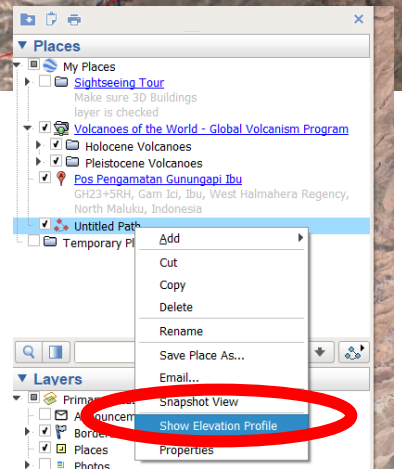
Okay, so how do we choose the H/L value?

PDC deposit fans are useful for choosing an ΔH/L cone/proximal hazard zone

- **Visualize volcano in 3-D:**
  - Find photos (Google image search, GVP)
  - Use Google Earth to get oblique views, fly around, and make a topographic profile.
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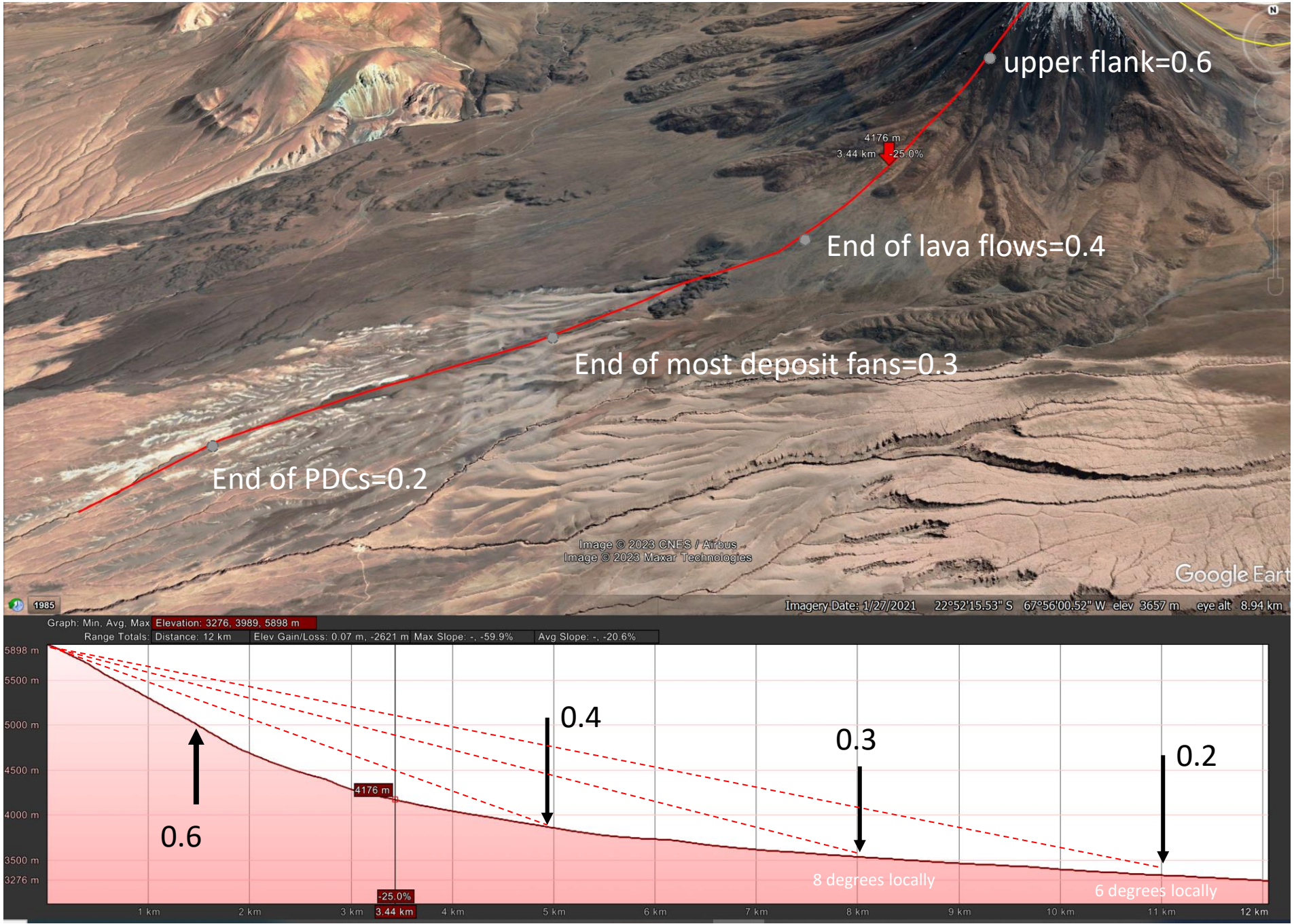
Visualize volcano in 3-D



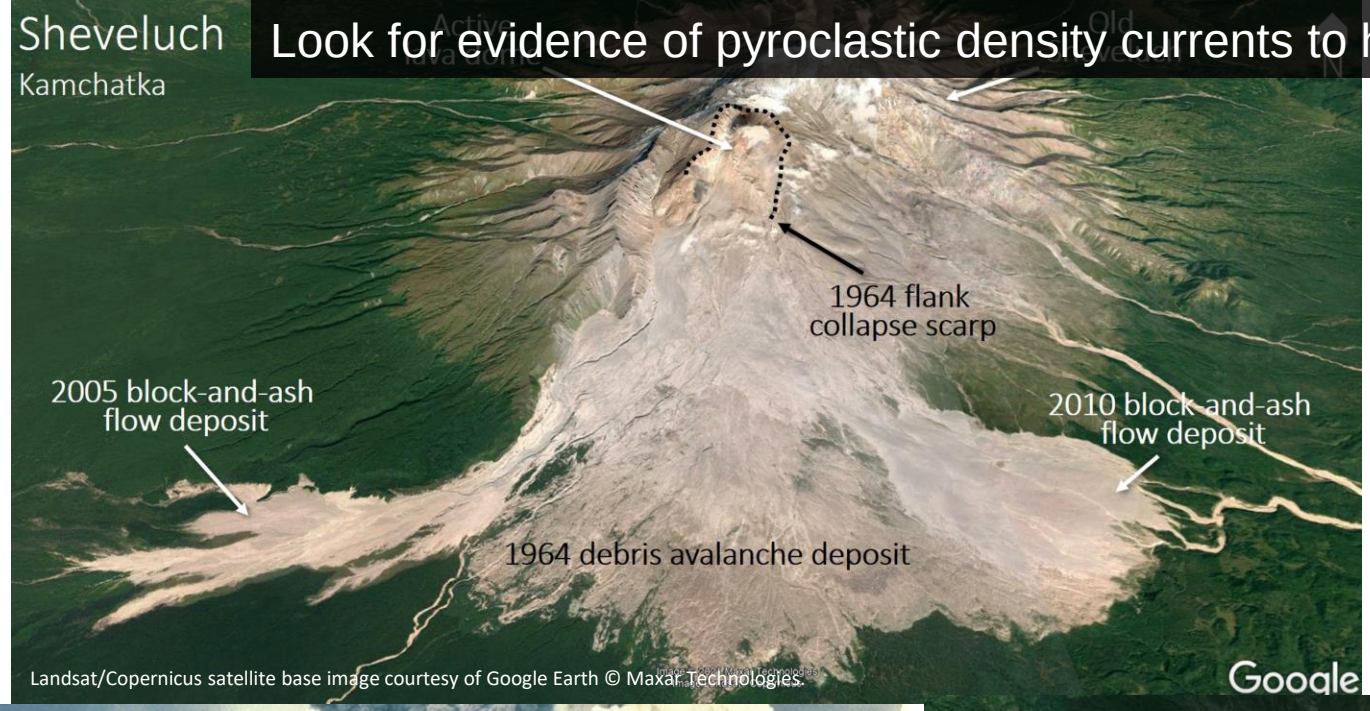


- Click on the “Add path” button
- Click on the map to make a path
- Name and save the path
- Right click on the path name under “Places”
- Choose “show elevation profile”
- Be careful: the slope values given in Google Earth are local, not H/L from the summit

Visualize volcano in 3-D



Look for evidence of pyroclastic density currents to help define an H/L ratio



Landsat/Copernicus satellite base image courtesy of Google Earth © Maxar Technologies

- **Look at the geomorphology**
- **Examine satellite imagery**

Google Search "GVP Mayon"



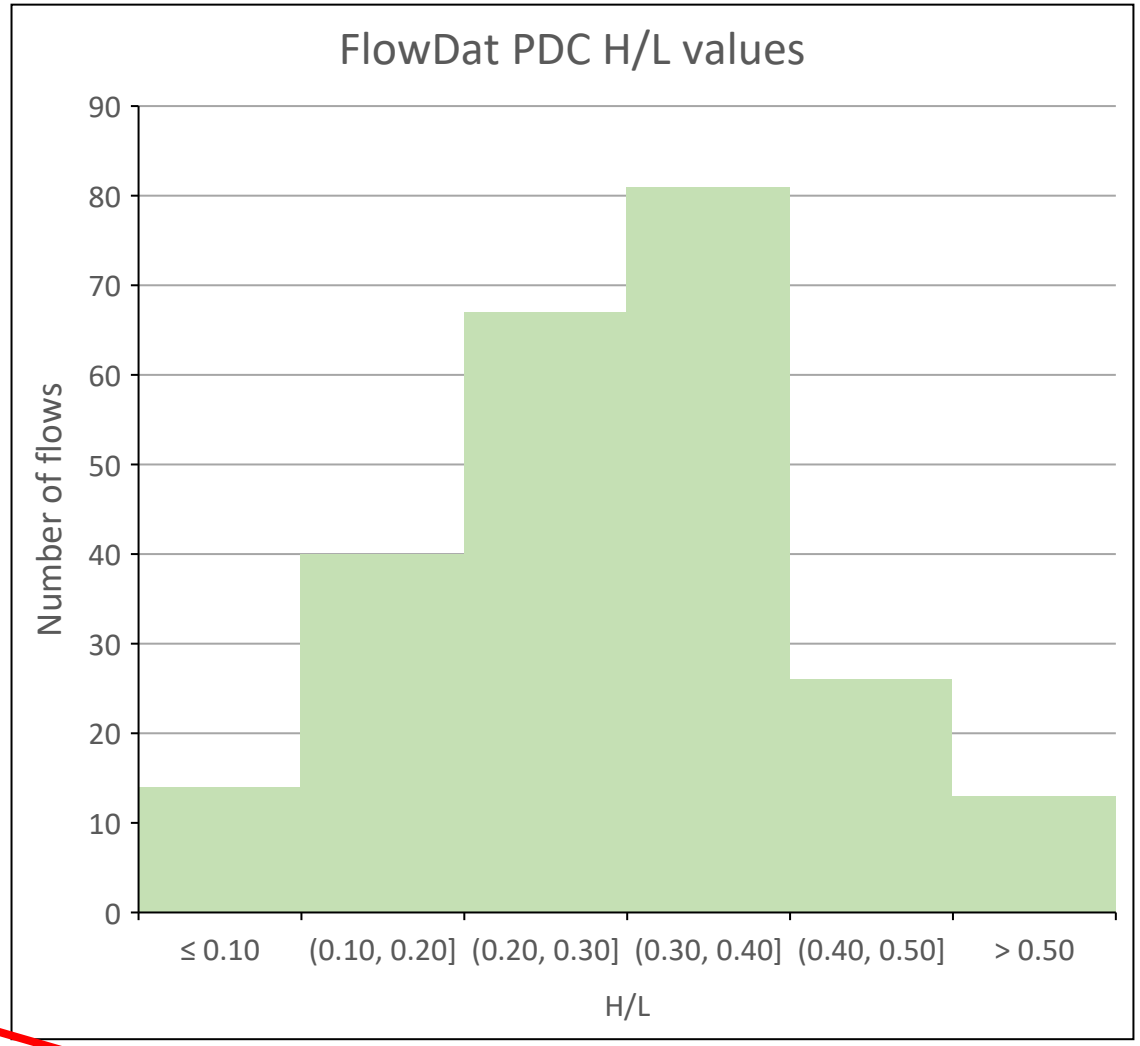
Mayon in 1984, USGS photo

Google Earth Mayon



Use global record of analog volcanoes (PDC H/L values)

- ΔH/L values range from ~0.04 for very large ignimbrites to 0.8 for very small dome-collapses or rockfalls
- Mean: 0.30
- Median: 0.30
- Most (80%) of ΔH/L values fall between 0.15 and 0.44



Start with H/L of 0.3 and refine

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Special Cases: not a nice pointy volcano?



Special Cases: not a nice pointy volcano?
Add elevation in LAHARZ



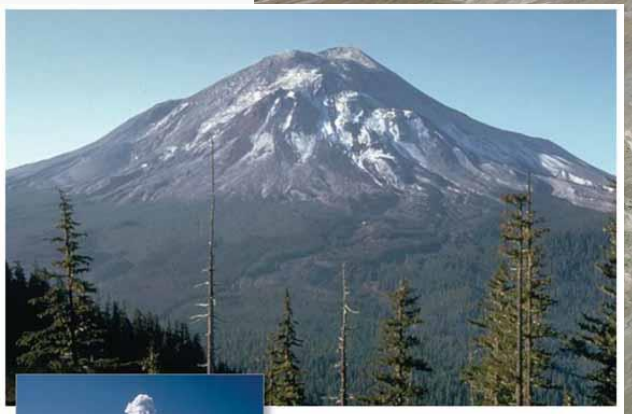
Maximum Elevation of the DEM

2950 meters

2539 meters along rim (2004 DEM)

2175 meters top of dome in 2004

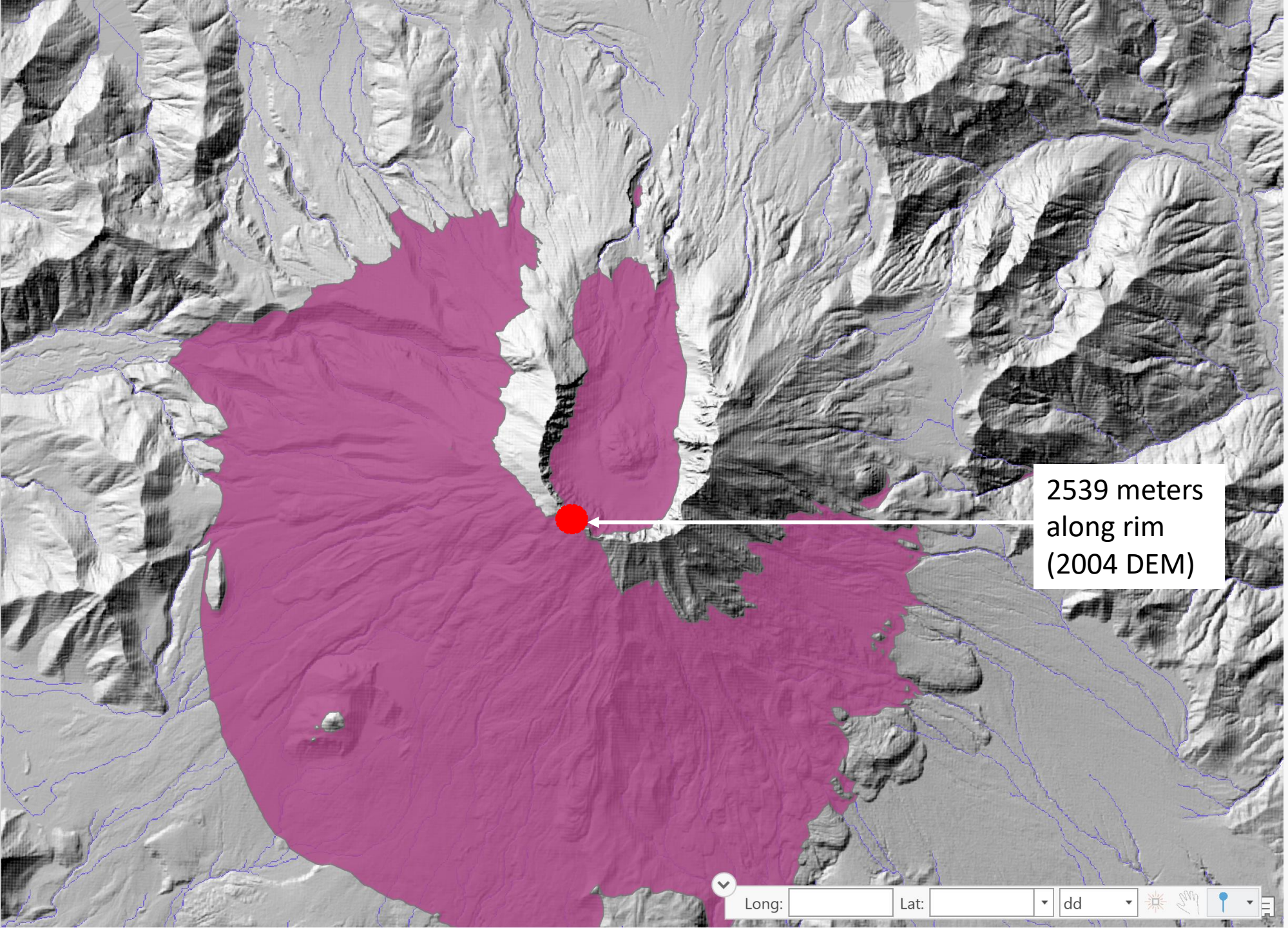
1960 meters crater floor





USGS photos

Image Landsat / Copernicus
Data LDEO-Columbia, NSF, NOAA
Google Earth
Imagery Date: 7/25/2021 lat 46.228684° lon -122.196775° elev 1377 m eye alt 2.35 km



This ΔH/L cone uses the highest point on the DEM, a point on the crater rim. This ΔH/L cone has holes and looks weird!

To-restablish the summit:
2950 m – 2175 m = add ~800 meters

2950 meters

2539 meters along rim (2004 DEM)

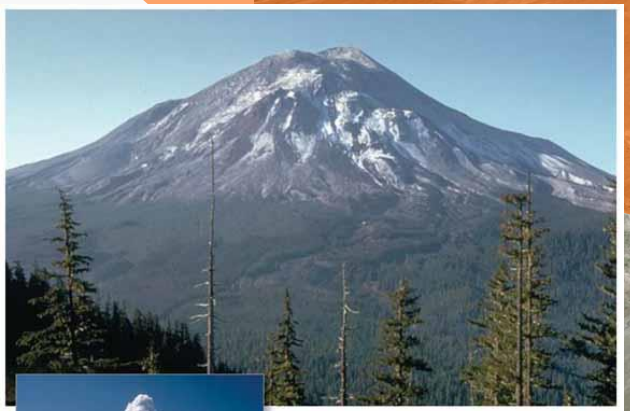
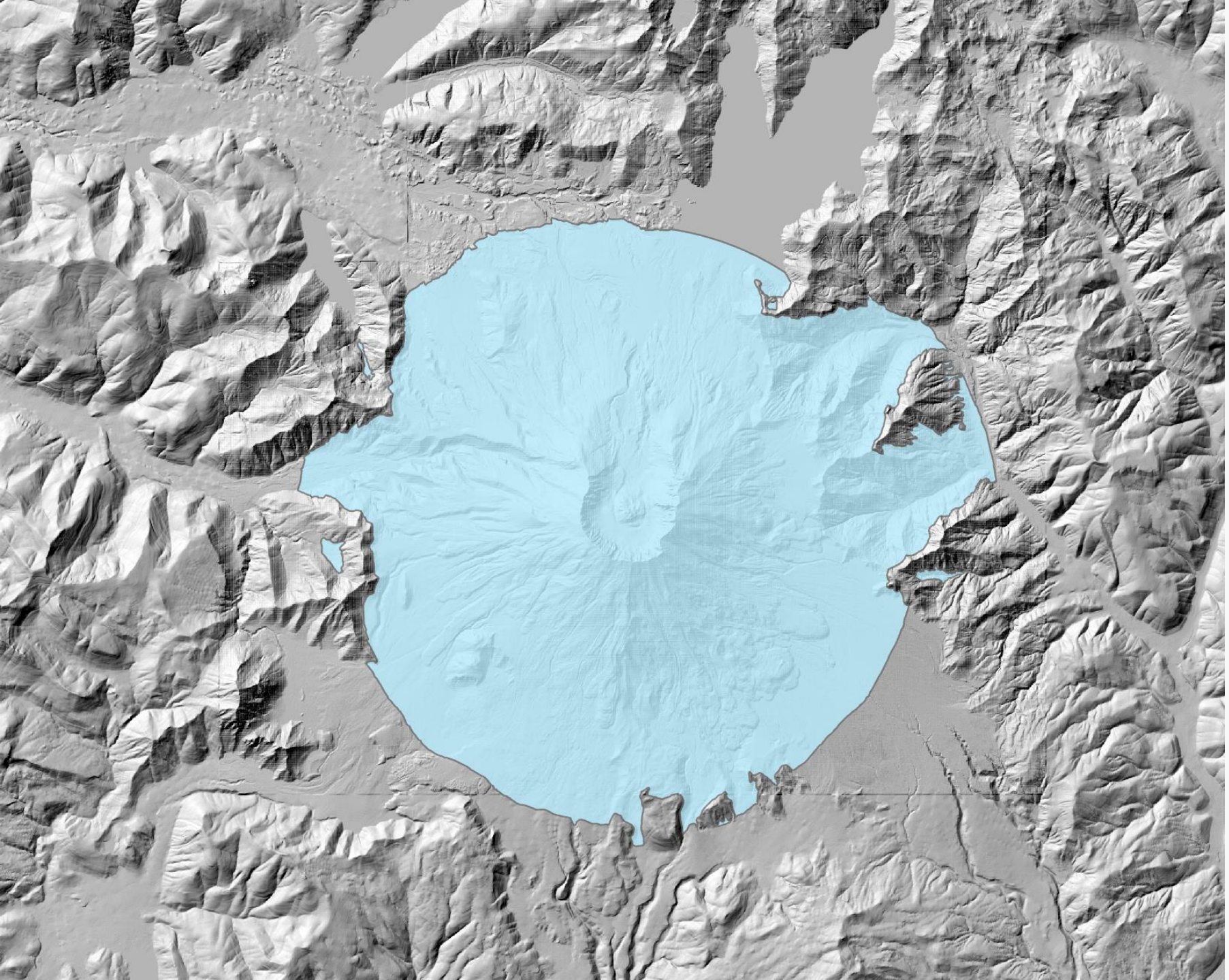
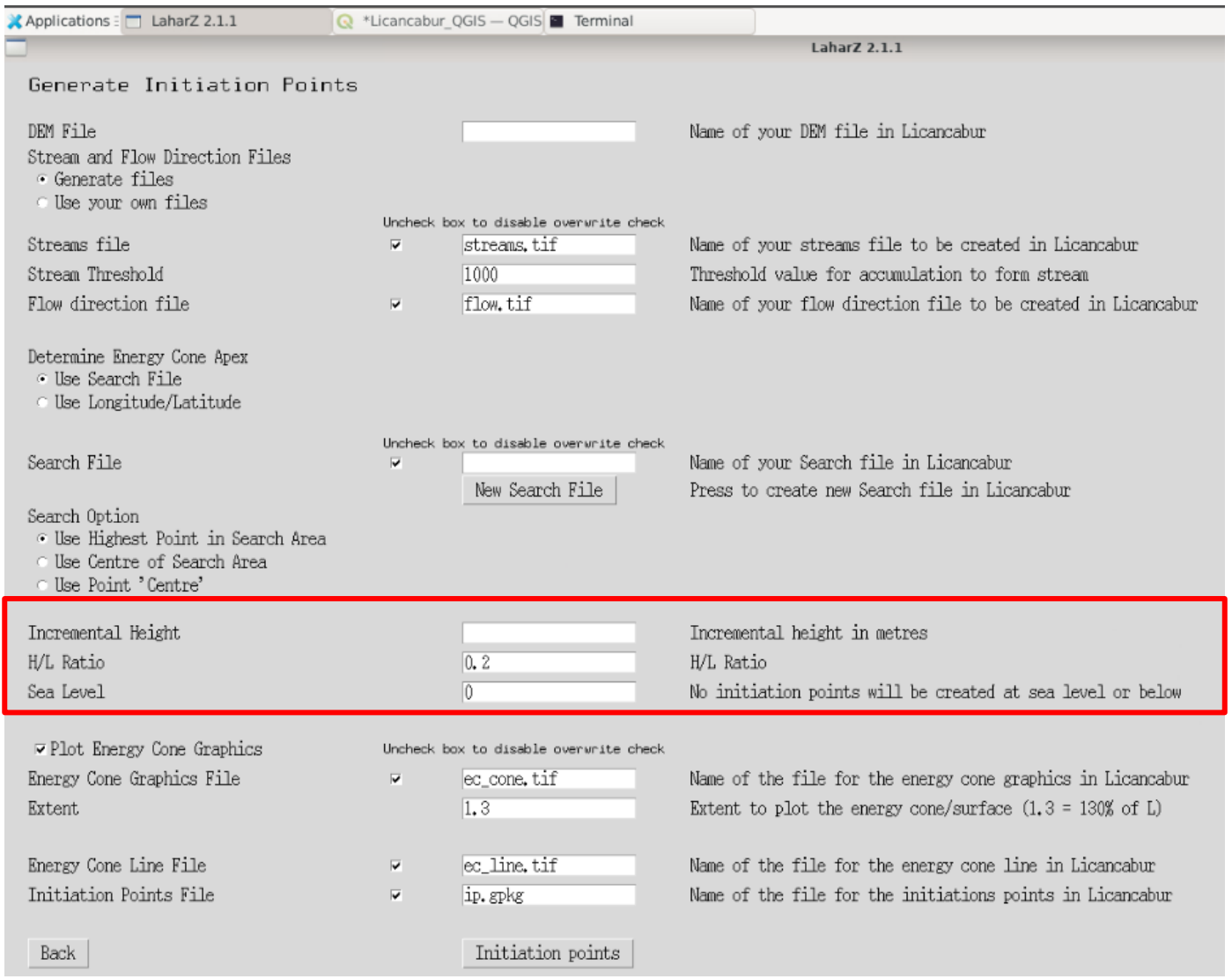


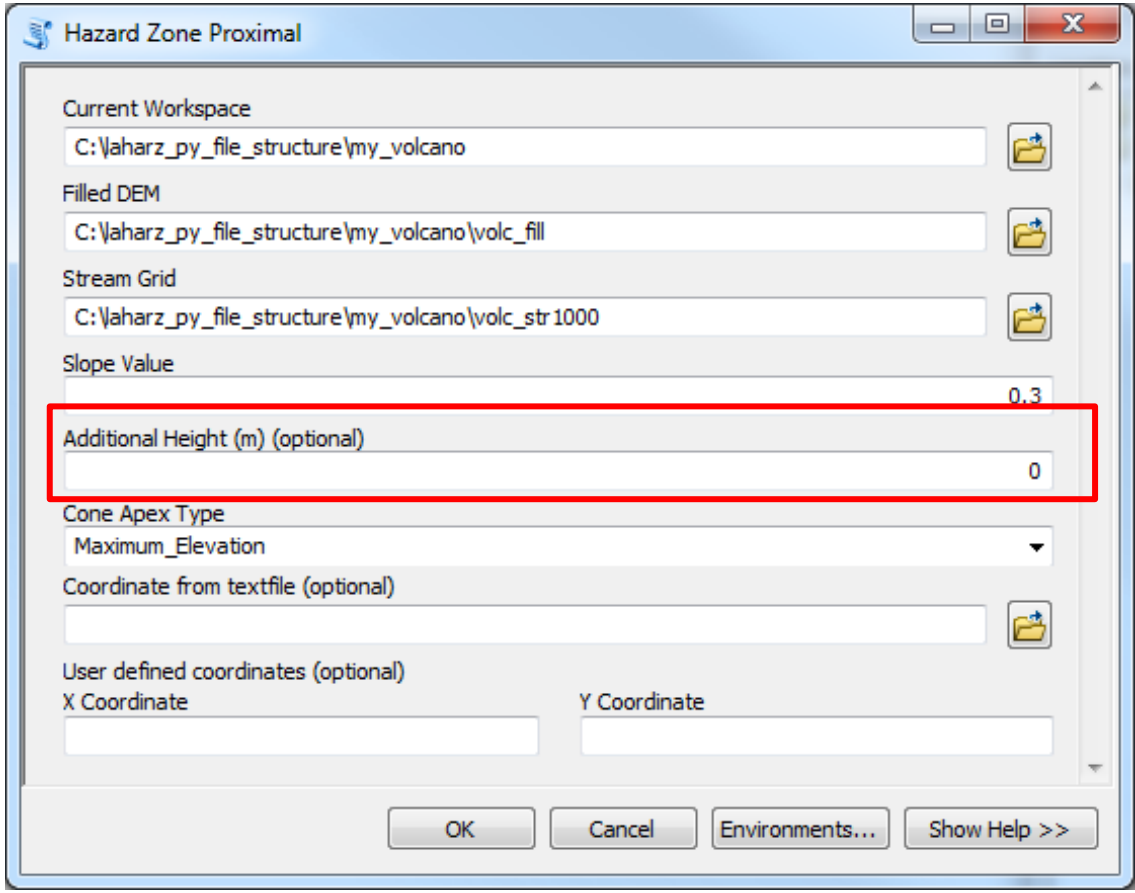


Image Landsat / Copernicus
Data LDEO-Columbia, NSF, NOAA
Google Earth
Imagery Date: 7/25/2021 lat 46.228684° lon -122.196775° elev 1377 m eye alt 2.35 km

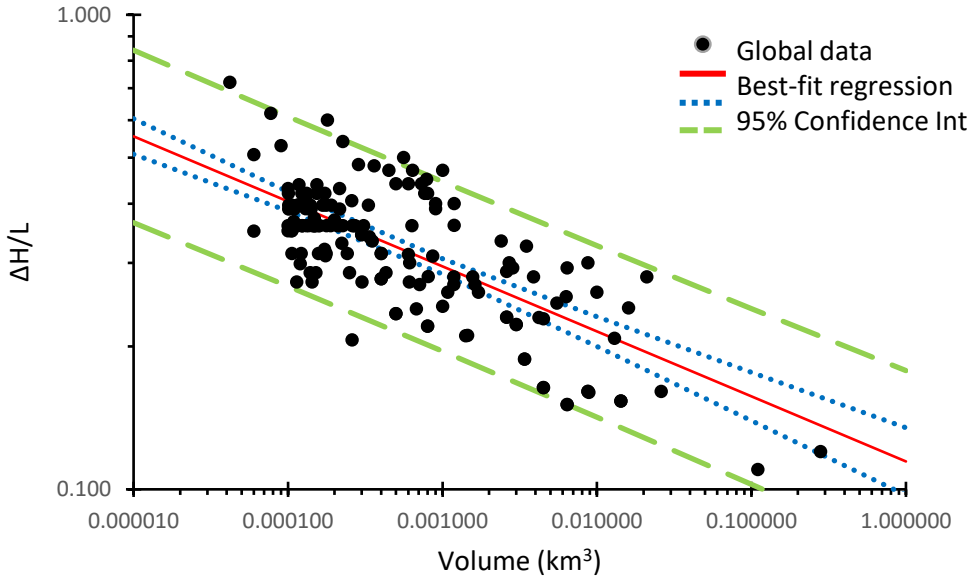


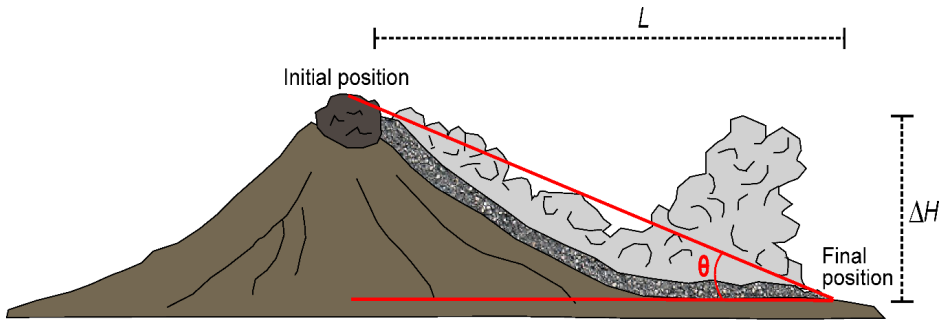
This ΔH/L cone adds ~800 m above the center of the crater. This ΔH/L cone has looks pretty good!

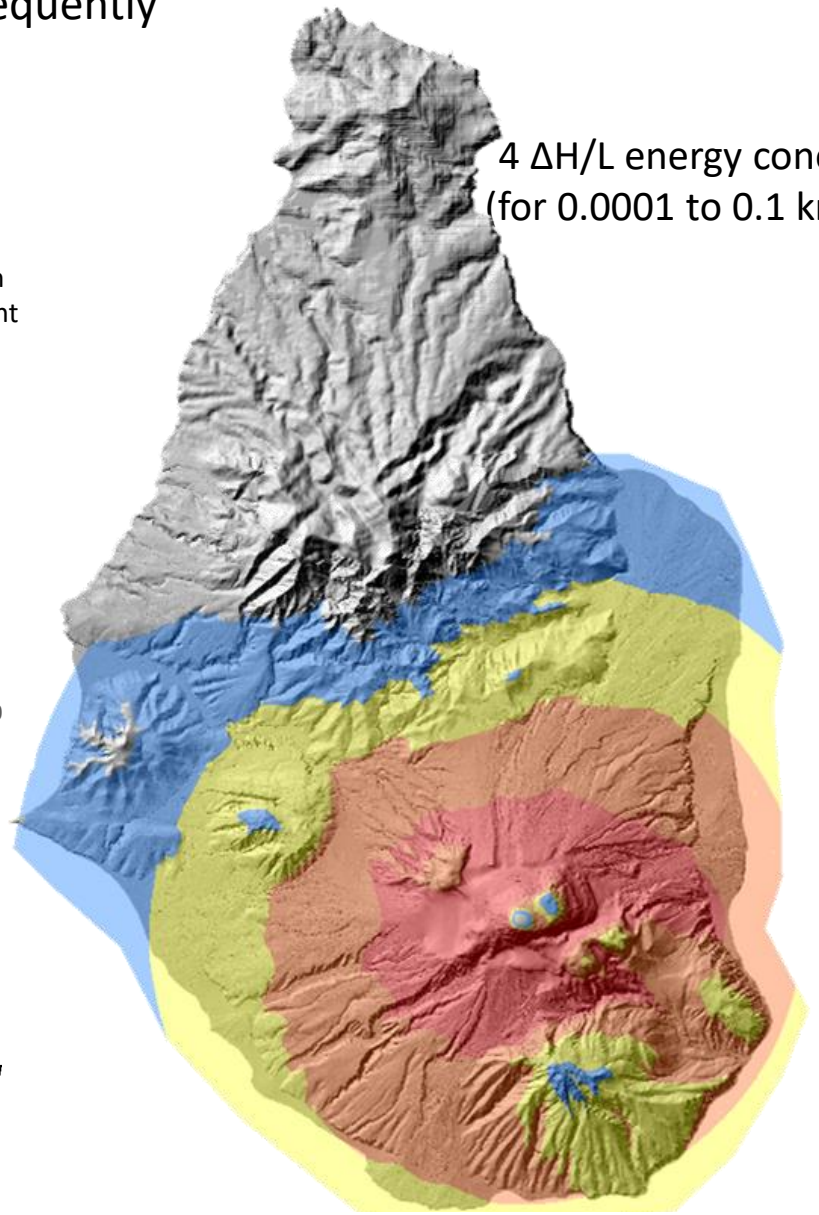

Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
Generate Initiation Points
DEM File
Name of your DEM file in Licancabur
Stream and Flow Direction Files
Generate files
Use your own files
Uncheck box to disable overwrite check
Streams file
streams.tif
Name of your streams file to be created in Licancabur
Stream Threshold
1000
Threshold value for accumulation to form stream
Flow direction file
flow.tif
Name of your flow direction file to be created in Licancabur
Determine Energy Cone Apex
Use Search File
Use Longitude/Latitude
Uncheck box to disable overwrite check
Search File
Name of your Search file in Licancabur
New Search File
Press to create new Search file in Licancabur
Search Option
Use Highest Point in Search Area
Use Centre of Search Area
Use Point 'Centre'
Incremental Height
Incremental height in metres
H/L Ratio
0.2
H/L Ratio
Sea Level
0
No initiation points will be created at sea level or below
Plot Energy Cone Graphics
Uncheck box to disable overwrite check
Energy Cone Graphics File
ec_cone.tif
Name of the file for the energy cone graphics in Licancabur
Extent
1.3
Extent to plot the energy cone/surface (1.3 = 130% of L)
Energy Cone Line File
ec_line.tif
Name of the file for the energy cone line in Licancabur
Initiation Points File
ip.gpkg
Name of the file for the initiations points in Licancabur
Back
Initiation points


Hazard Zone Proximal
Current Workspace
C:\laharz_py_file_structure\my_volcano
Filled DEM
C:\laharz_py_file_structure\my_volcano\volc_fill
Stream Grid
C:\laharz_py_file_structure\my_volcano\volc_str1000
Slope Value
0.3
Additional Height (m) (optional)
0
Cone Apex Type
Maximum_Elevation
Coordinate from textfile (optional)
User defined coordinates (optional)
X Coordinate
Y Coordinate
OK
Cancel
Environments...
Show Help >>

Relationship between PDC volume and H/L is frequently used in energy cone modeling of PDCs
(we'll talk about this more later!)







4 ΔH/L energy cones
(for 0.0001 to 0.1 km^3)

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